

NB13 - Kapsamli Ablasyon Deneyi

No Feature Engineering | OHE | XGBoost + LightGBM

Tarih: 2026-05-12 11:03

1. Pipeline & Veri

Veri: YARISMA_TRAIN_MASTER.csv (2931 satir, 353 sutun)

NaN>50% drop: 165 sutun cikarildi -> kalan 186

Encoding: OHE (CAT_+AA_), mean imputation (AL_+EK_) - fit sadece train

Model: XGBoost + LightGBM, 3-fold CV grid search (12 combo)

Split: stratified hold-out %20 (train=2344, test=587)

Senaryo sayisi: 13 ablasyon + 3 missing-mask

2. Ablasyon Sonuclari (Hold-out)

Model: XGBoost

Senaryo	n_feat	CV F1	F1	dF1	AUC	dAUC	AUC-PR	MCC	Recall
Baseline	280	0.8876	0.8985	+0.0000	0.8474	+0.0000	0.9168	0.5600	0.9674
-CAT_all	231	0.8838	0.8997	-0.0012	0.8438	+0.0036	0.9093	0.5654	0.9698
-CAT_pop	249	0.8853	0.9025	-0.0040	0.8387	+0.0087	0.9098	0.5865	0.9581
-CAT_geno	262	0.8883	0.8978	+0.0007	0.8441	+0.0033	0.9134	0.5602	0.9605
-CAT_dup	268	0.8902	0.8990	-0.0005	0.8455	+0.0019	0.9149	0.5653	0.9628
-AA	229	0.8854	0.8889	+0.0096	0.8199	+0.0275	0.8999	0.5193	0.9488
-EK	271	0.8805	0.8845	+0.0140	0.8192	+0.0282	0.9128	0.5077	0.9349
-AL_safe	109	0.8720	0.8710	+0.0275	0.7883	+0.0591	0.8967	0.3815	0.9814
-AL_high_miss	280	0.8876	0.8985	+0.0000	0.8474	+0.0000	0.9168	0.5600	0.9674
-AL_miss_leak	280	0.8876	0.8985	+0.0000	0.8474	+0.0000	0.9168	0.5600	0.9674
-AL_low	269	0.8792	0.8901	+0.0084	0.8403	+0.0071	0.9254	0.5116	0.9698
-AL_high	120	0.8744	0.8845	+0.0140	0.8039	+0.0435	0.8953	0.4736	0.9791
-AL_all	109	0.8720	0.8710	+0.0275	0.7883	+0.0591	0.8967	0.3815	0.9814

Model: LightGBM

Senaryo	n_feat	CV F1	F1	dF1	AUC	dAUC	AUC-PR	MCC	Recall
Baseline	280	0.8859	0.8973	+0.0000	0.8344	+0.0000	0.9108	0.5547	0.9651
-CAT_all	231	0.8847	0.8950	+0.0023	0.8293	+0.0051	0.9128	0.5510	0.9512
-CAT_pop	249	0.8843	0.9048	-0.0075	0.8379	-0.0035	0.9110	0.5916	0.9721
-CAT_geno	262	0.8860	0.8975	-0.0002	0.8340	+0.0004	0.9084	0.5679	0.9465
-CAT_dup	268	0.8858	0.8991	-0.0018	0.8362	-0.0018	0.9127	0.5607	0.9744
-AA	229	0.8863	0.8942	+0.0031	0.8118	+0.0226	0.8925	0.5387	0.9628
-EK	271	0.8799	0.8825	+0.0148	0.8050	+0.0294	0.8993	0.4724	0.9605
-AL_safe	109	0.8701	0.8763	+0.0210	0.7942	+0.0402	0.8915	0.4173	0.9884
-AL_high_miss	280	0.8859	0.8973	+0.0000	0.8344	+0.0000	0.9108	0.5547	0.9651
-AL_miss_leak	280	0.8859	0.8973	+0.0000	0.8344	+0.0000	0.9108	0.5547	0.9651
-AL_low	269	0.8822	0.8908	+0.0065	0.8383	-0.0039	0.9205	0.5104	0.9860
-AL_high	120	0.8755	0.8833	+0.0140	0.7939	+0.0405	0.8789	0.4672	0.9767
-AL_all	109	0.8701	0.8763	+0.0210	0.7942	+0.0402	0.8915	0.4173	0.9884

3. Missing-Mask Ablasyonu

Senaryo	Model	n_mask	F1	AUC-ROC	MCC	Recall
Baseline_nomask	XGBoost	0	0.8985	0.8474	0.5600	0.9674
Baseline_nomask	LightGBM	0	0.8973	0.8344	0.5547	0.9651
+Mask_all	XGBoost	180	0.8974	0.8451	0.5484	0.9860
+Mask_all	LightGBM	180	0.8966	0.8291	0.5551	0.9581
+Mask_highrisk	XGBoost	0	0.8985	0.8474	0.5600	0.9674
+Mask_highrisk	LightGBM	0	0.8973	0.8344	0.5547	0.9651

4. Panel Transfer Sonuclari

Panel	Model	F1	AUC-ROC	AUC-PR	Precision	Recall	MCC
CFTR	XGBoost	0.9348	0.9492	0.9884	0.9149	0.9556	0.6249
KANSER	XGBoost	0.8930	0.9019	0.9436	0.8193	0.9813	0.6090
PAH	XGBoost	0.9309	0.7852	0.9264	0.8886	0.9774	0.4915
CFTR	LightGBM	0.9457	0.9386	0.9853	0.9255	0.9667	0.6888
KANSER	LightGBM	0.8956	0.9122	0.9500	0.8160	0.9925	0.6213
PAH	LightGBM	0.9344	0.8406	0.9537	0.8870	0.9871	0.5143

5. Bulgular

XGBoost Baseline: F1=0.8985 AUC=0.8474 MCC=0.5600

En kritik: -AL_safe (F1 dusus: +0.0275)

En az etkili: -CAT_pop (F1 degisim: -0.0040)

LightGBM Baseline: F1=0.8973 AUC=0.8344 MCC=0.5547

En kritik: -AL_safe (F1 dusus: +0.0210)

En az etkili: -CAT_pop (F1 degisim: -0.0075)

Yorum: Pozitif dF1 = grubun cikarilmasi F1 dusurdu (grup faydali). Negatif dF1 = grup cikarildiginda model iyilesti (gurultu/leakage riski). En yuksek dF1 olan grup, feature engineering ile zenginlestirilmeye oncelikli aday. Missing-mask katkisi pozitifse eksiklik deseni etiketle koreleli (leakage riski).

Fig1 F1Drop Comparison

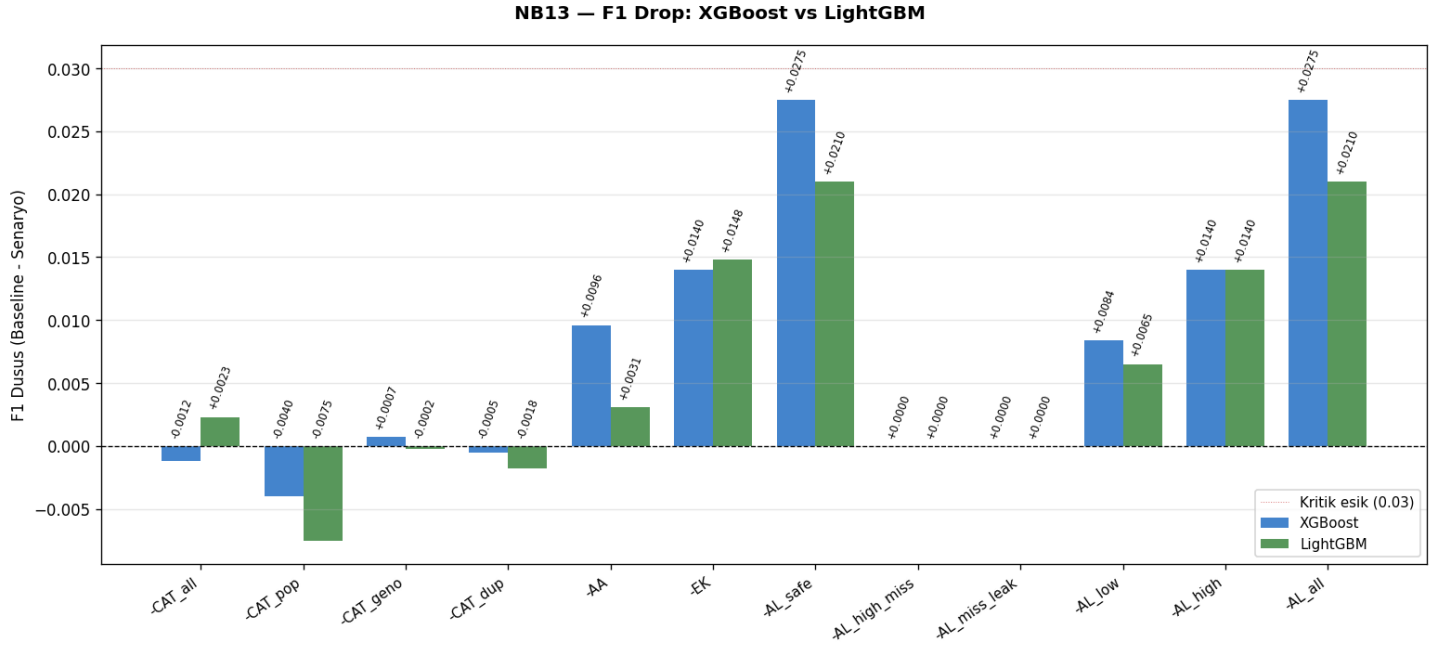


Fig2 Metrics Xgb

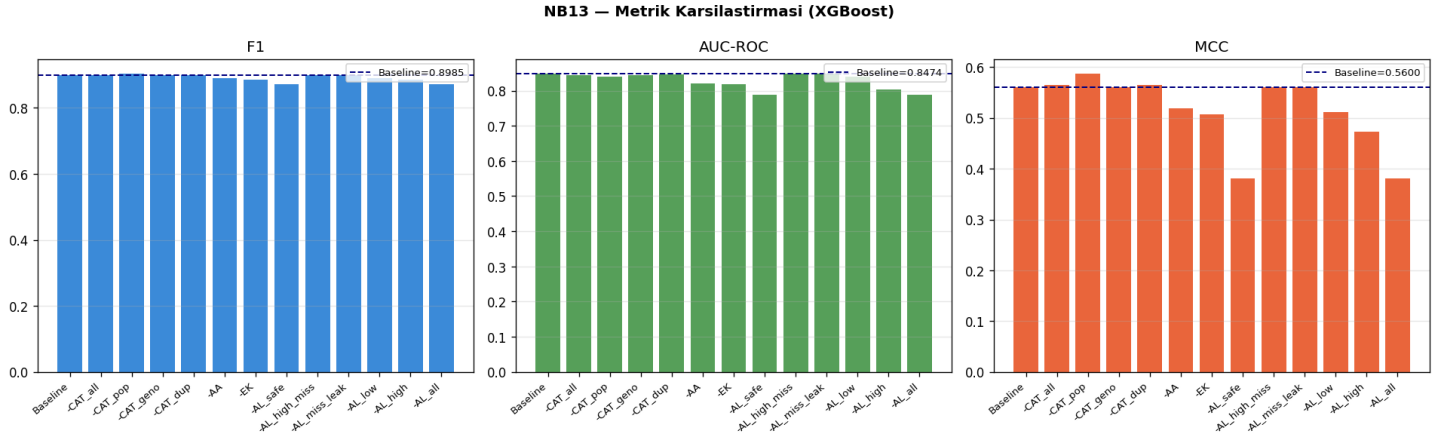


Fig3 Panel Heatmap

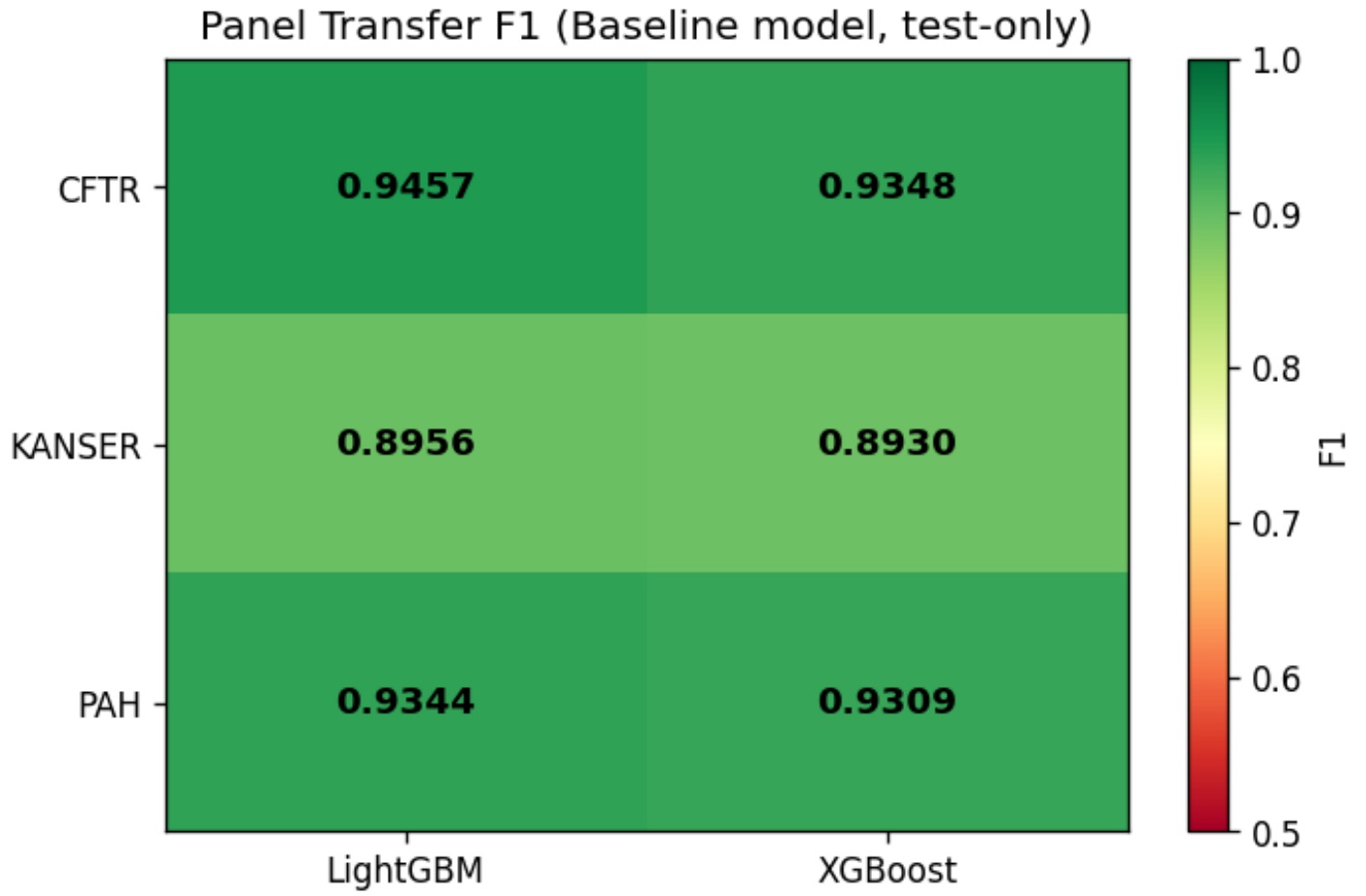


Fig4 Missing Mask

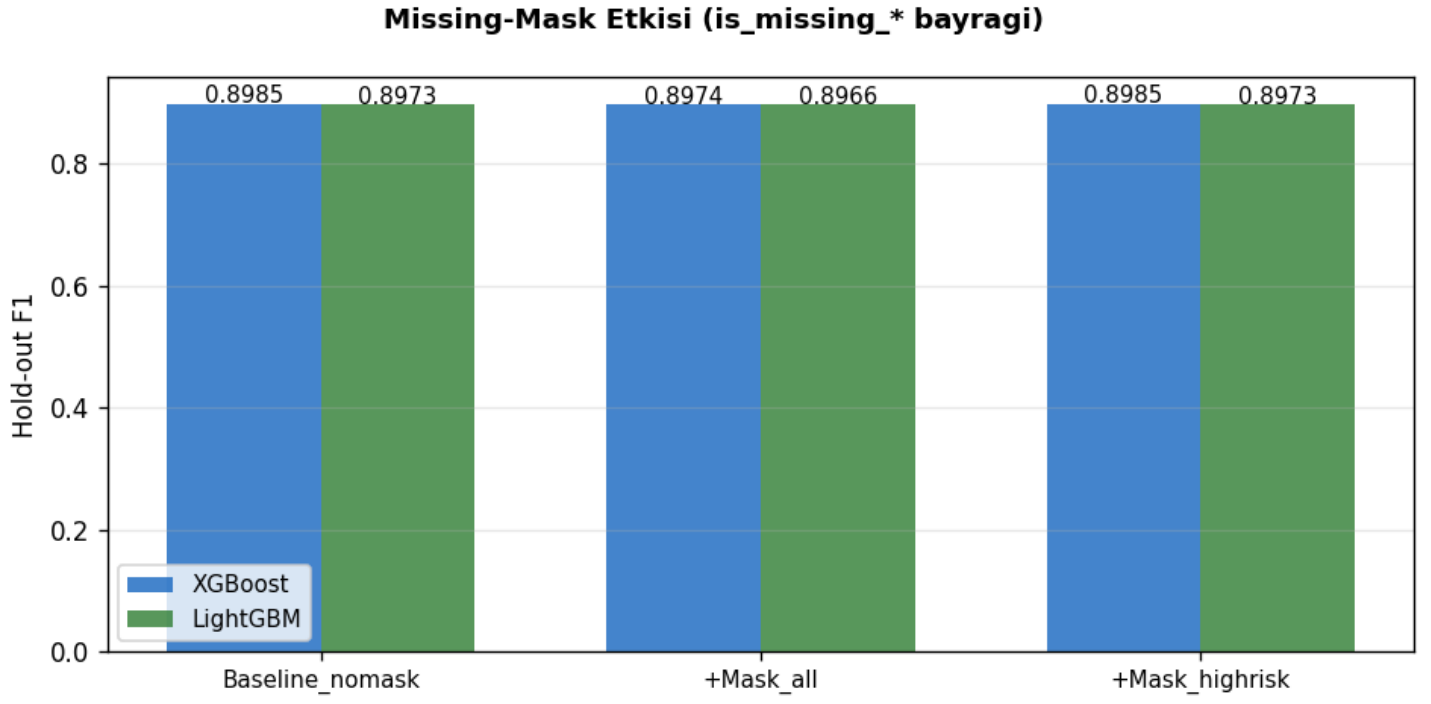


Fig5 Aucpr Drop

